Update Conference
PRAGUE 2026

# Agentic AI on Microsoft Foundry: From Function Calling to Multi-Agent Systems



**Cédric Mendelin**

Software Developer | Team Lead | isolutions AG

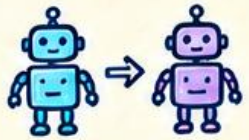
# THE CATALOG OF DESIGN PATTERNS

## CREATIONAL PATTERNS


Factory Method


Abstract Factory


Builder


Prototype

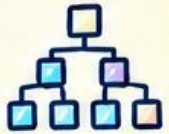

Singleton

CREATE OBJECTS FLEXIBLY

## STRUCTURAL PATTERNS

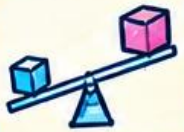

Adapter

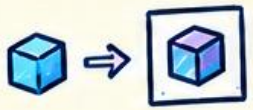

Bridge


Composite


Decorator


Facade


Flyweight


Proxy

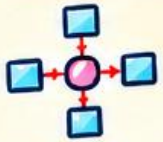
ASSEMBLE FLEXIBLE STRUCTURES

## BEHAVIORAL PATTERNS


Chain of Responsibility

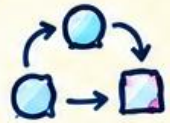

Command


Iterator

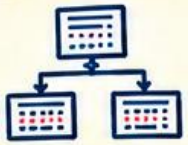

Mediator

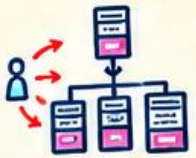

Memento


Observer

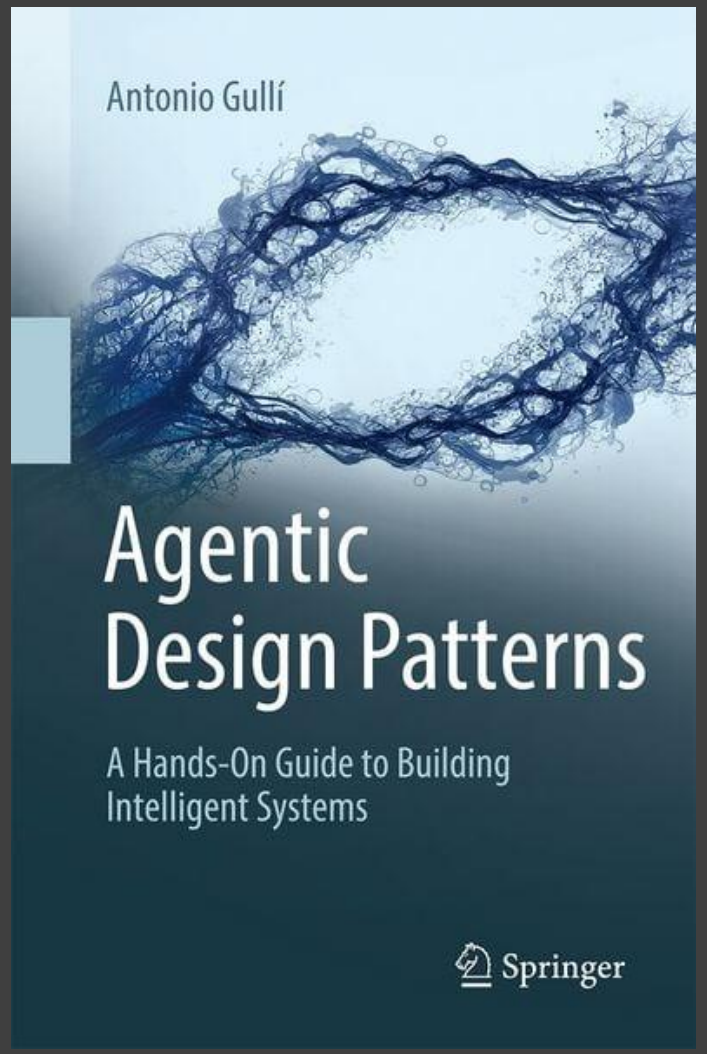

State

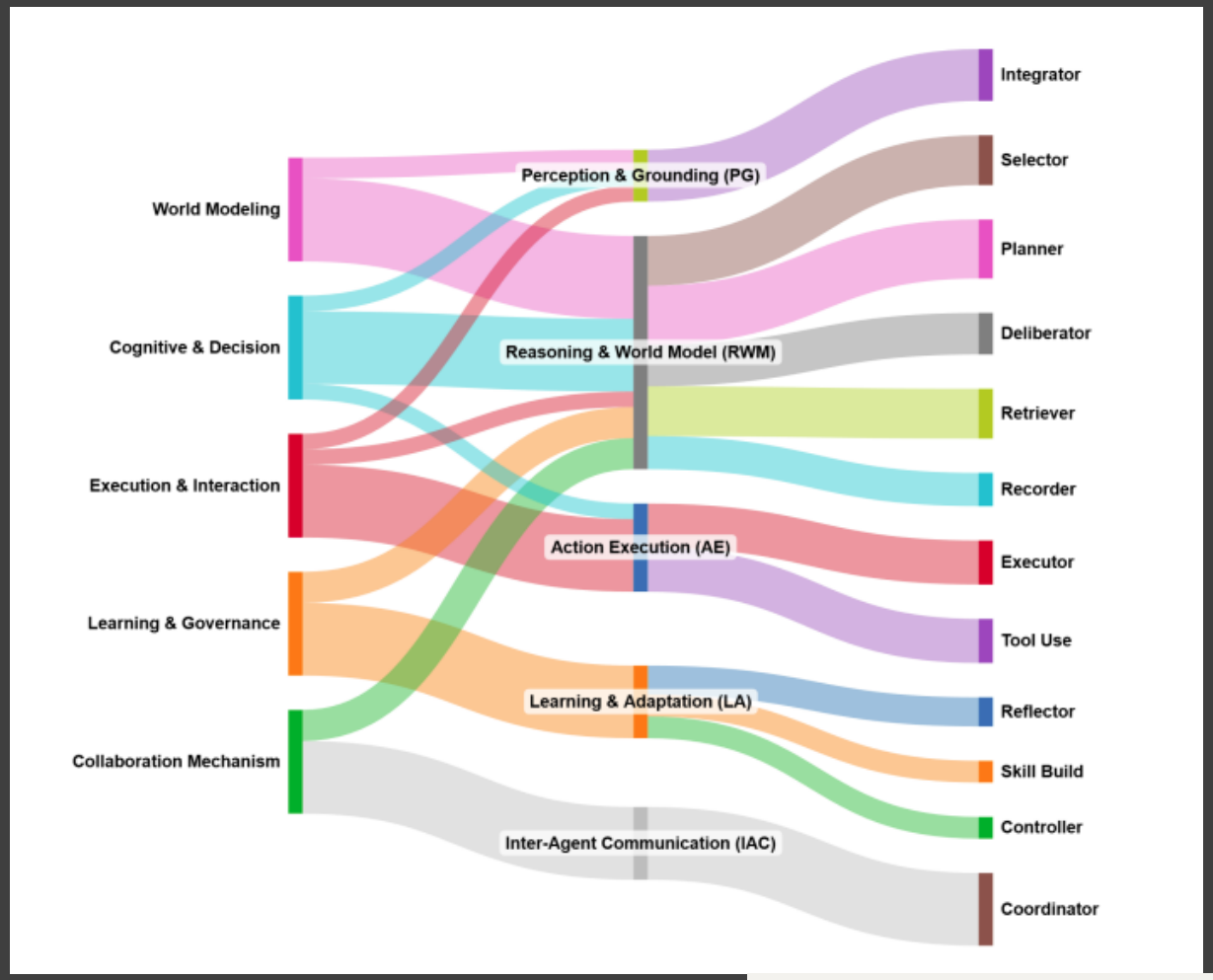

Strategy

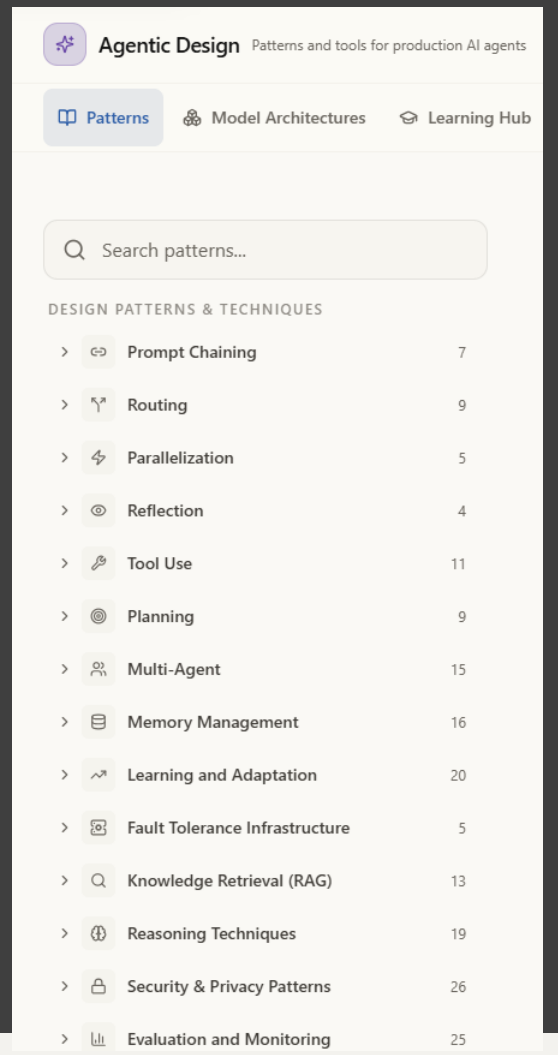

Template Method

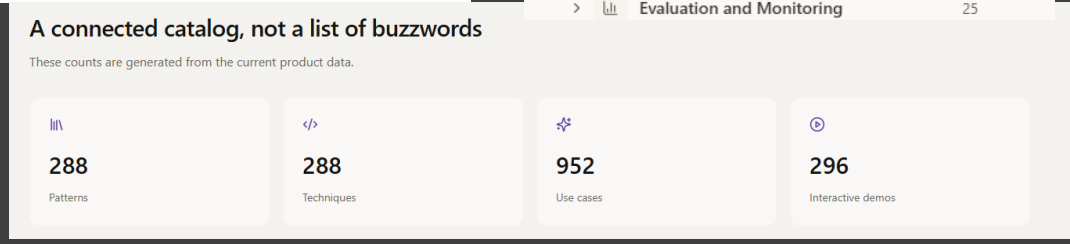

Visitor

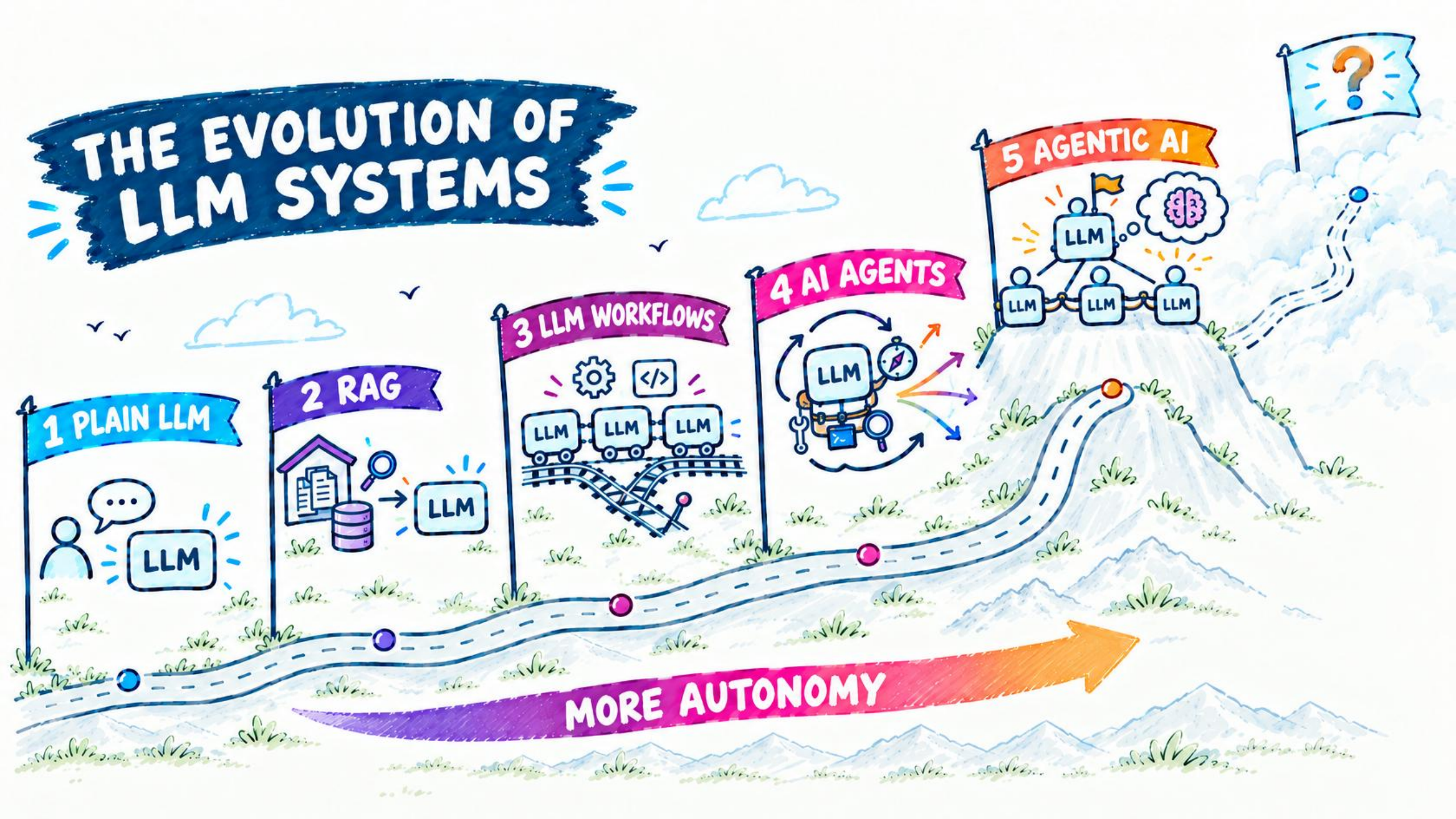
ASSIGN RESPONSIBILITIES


Antonio Gullí
Agentic
Design Patterns
A Hands-On Guide to Building
Intelligent Systems
Springer


World Modeling
Cognitive & Decision
Execution & Interaction
Learning & Governance
Collaboration Mechanism
Perception & Grounding (PG)
Reasoning & World Model (RWM)
Action Execution (AE)
Learning & Adaptation (LA)
Inter-Agent Communication (IAC)
Integrator
Selector
Planner
Deliberator
Retriever
Recorder
Executor
Tool Use
Reflector
Skill Build
Controller
Coordinator


Agentic Design
Patterns and tools for production AI agents
Patterns
Model Architectures
Learning Hub
Search patterns...
DESIGN PATTERNS & TECHNIQUES
Prompt Chaining 7
Routing 9
Parallelization 5
Reflection 4
Tool Use 11
Planning 9
Multi-Agent 15
Memory Management 16
Learning and Adaptation 20
Fault Tolerance Infrastructure 5
Knowledge Retrieval (RAG) 13
Reasoning Techniques 19
Security & Privacy Patterns 26
Evaluation and Monitoring 25


A connected catalog, not a list of buzzwords
These counts are generated from the current product data.
288
Patterns
288
Techniques
952
Use cases
296
Interactive demos


THE EVOLUTION OF LLM SYSTEMS
1 PLAIN LLM
LLM
2 RAG
LLM
3 LLM WORKFLOWS
LLM
LLM
LLM
4 AI AGENTS
LLM
5 AGENTIC AI
LLM
LLM
LLM
LLM
?
MORE AUTONOMY

# CÉDRIC MENDELIN

Software Developer & Team Lead

isolutions




Microsoft® MVP Most Valuable Professional

# MOTIVATION

) one of
e played
at risks,
cut into

g risky,

ies) di-
efined
ed to

**dictatorial** /ˌdɪktəˈtɔːrɪəl/ like a dictator. **2** overbearing. **orially** *adv.* [Latin: related TATOR]

**diction** /ˈdɪkʃ(ə)n/ *n.* manner ciation in speaking or singing *dictio* from *dico dict-* say]

**dictionary** /ˈdɪkʃənərɪ/ *n.* (pl book listing (usu. alphabetica explaining the words of a lan giving corresponding words in language. **2** reference book e

# WHAT IS THE CURRENT DATE?


1+1=3

# Which sessions should I attend at Update Conference 2026 if I'm interested in AI and .NET?

**DAY 1** November 12 | **DAY 2** November 13

| | HALL A | HALL B | HALL C | HALL D |
|---|---|---|---|---|
| 08:30 | Registration & Breakfast | | | |
| 08:45 | | | | |
| 09:00 | | | | |
| 09:15 | #1 EN Intermediate 60 min<br>After the AI Hype – What's Real, and What's Next<br>Richard Campbell SHOW MORE | #2 EN Introduction 60 min<br>What's new in .NET 11<br>Johan Smarius SHOW MORE | #3 EN Intermediate 60 min<br>Five things every developer should know about building mission-critical systems<br>Loek Duys SHOW MORE | #4 EN Introduction 60 min<br>What Makes Good UI Work: Lessons from Cognitive Psychology<br>Keren Kenzi SHOW MORE |
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| 16:00 | | | | |


USERS
WORLD
AI AGENT
TOOLS
LLM
MEMORY
ACTION

# Which sessions should I attend at Update Conference 2026 if I'm interested in AI and .NET?

Check my proposal agenda.

Fix overlaps and unsupported choices.

I'll miss Christian's session about EU Cyber Resilience Act. Where else can I hear him?

# AGENT COMPLEXITY LEVELS

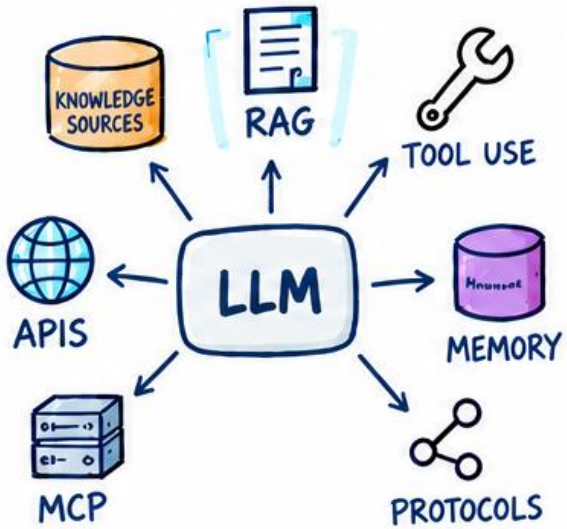
## 0 REASON

A REASONING ENGINE




## 1 CONNECT

A REASONING ENGINE
WITH A CONNECTION TO THE WORLD



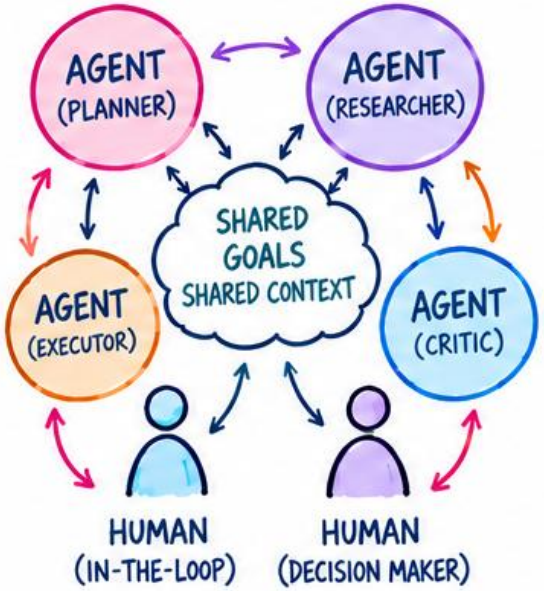

## 2 STRATEGIZE

A STRATEGIC PROBLEM SOLVER




## 3 COLLABORATE

A TEAM OF AGENTS AND HUMANS
WORKING TOGETHER






# AGENTIC DESIGN PATTERN FAMILIES

(APPLY ACROSS ALL LEVELS)

## FLOW

- Prompt Chaining
- Routing
- Parallelization
- Planning
- Prioritization
- Decision Models

ORCHESTRATE WORK EFFECTIVELY

## REASON & IMPROVE

- Reflection
- Learning & Adaptation
- Goal Setting & Monitoring
- Reasoning
- Exploration & Discovery

GET BETTER OVER TIME

## KNOWLEDGE & ACTION

- Tool Use
- Memory
- MCP
- RAG

BRIDGE IDEAS AND THE REAL WORLD

## COLLABORATION

- Multi-Agent Collaboration
- Human-in-the-Loop
- A2A




ACHIEVE MORE TOGETHER

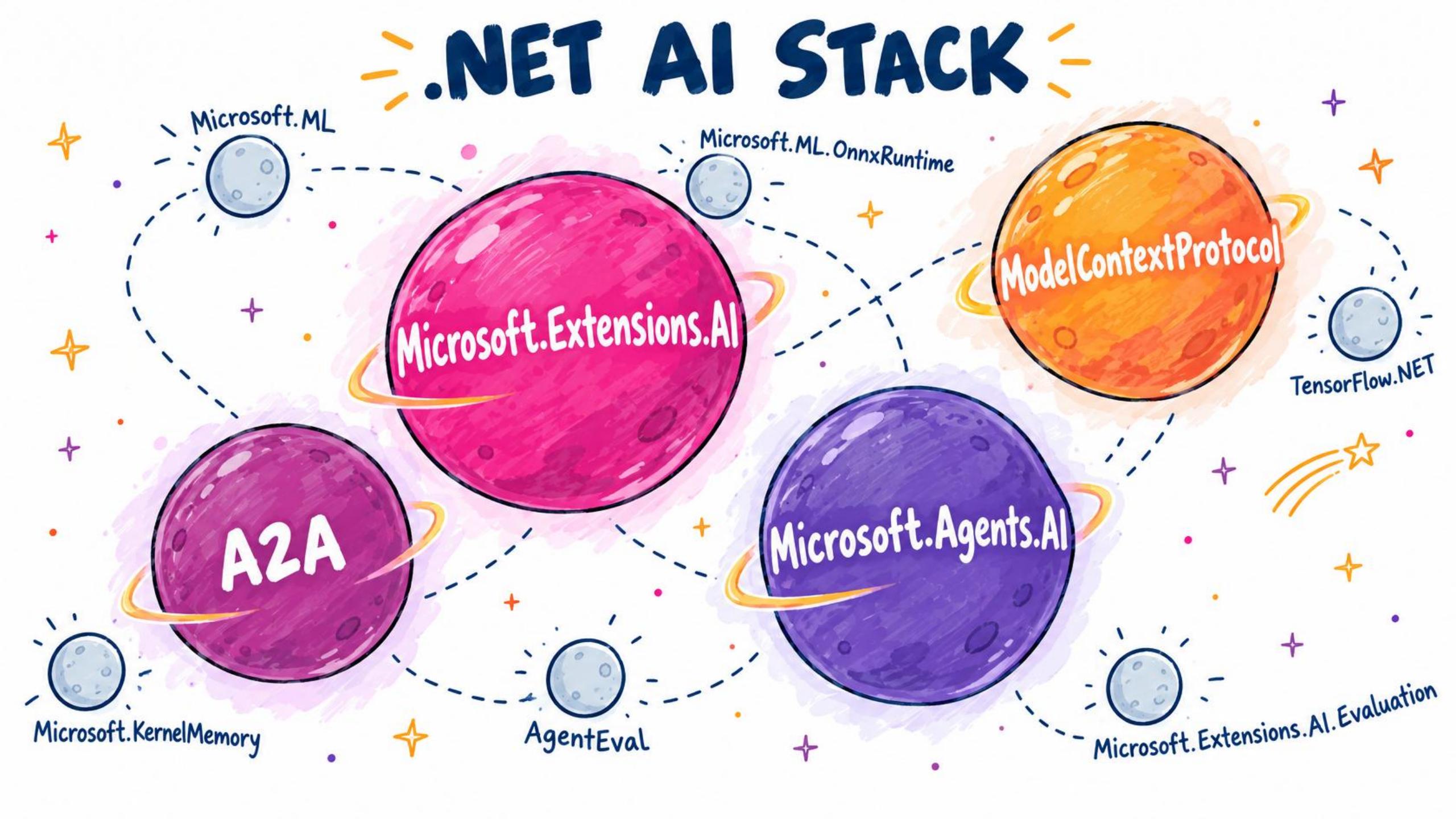
## TRUST & OPERATIONS

- Exception Handling
- Resource Optimization
- Guardrails
- Evaluation & Monitoring
- Agent Harness

A SAFE, RELIABLE AND SUSTAINABLE FOUNDATION FOR AGENTS AT ANY SCALE

# C# ❤️ AI




.NET AI Stack
Microsoft.ML
Microsoft.ML.OnnxRuntime
Microsoft.Extensions.AI
ModelContextProtocol
TensorFlow.NET
A2A
Microsoft.Agents.AI
Microsoft.KernelMemory
AgentEval
Microsoft.Extensions.AI.Evaluation

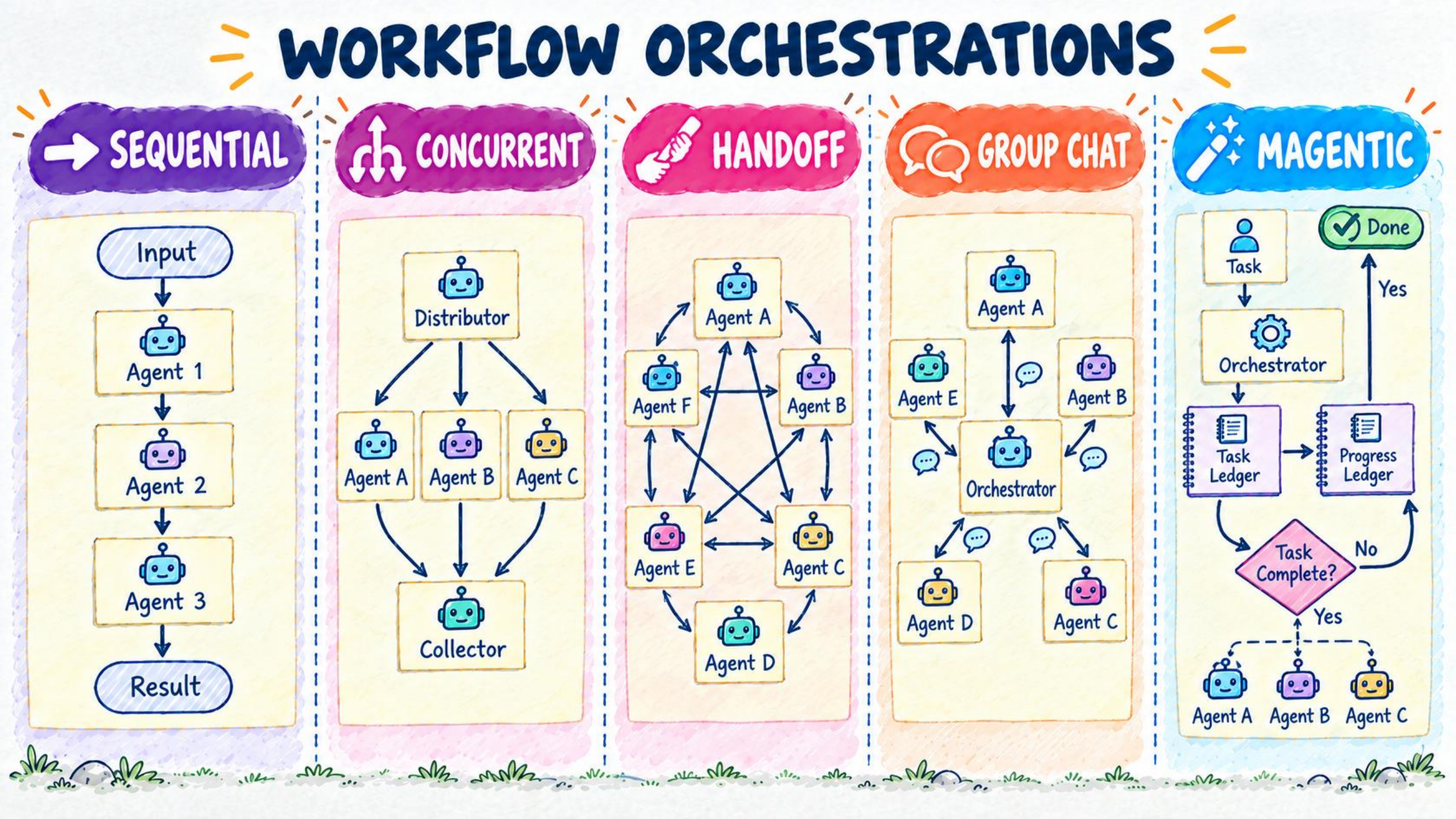
```
// 1. settings
var apikey = "";
var endpoint = "";
var model = "";

// 2. initialize MEAI client
var client = new OpenAIClient(
    new ApiKeyCredential(apikey),
    new OpenAIClientOptions { Endpoint = endpoint  });
var chatClient = client.GetChatClient(model).AsIChatClient();

// 3. create MAF Agent
var agent = new ChatClientAgent(
    chatClient,
    name: "PragueAssistant",
    instructions: """
        You are a concise, friendly assistant for Update Conference in Prague.
        For Prague weather questions, call GetPragueWeather.
        """,
    tools: [AIFunctionFactory.Create(() => "Prague is cloudy, 6 C." , name: "GetPragueWeather")]
    );

// 4. execute
var response = await agent.RunAsync("What is the weather like in Prague today?");
```

# WORKFLOW ORCHESTRATIONS



**DAY 1** November 12 | **DAY 2** November 13

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```
// 1. create executors
Func<string, string> getProfile = attendee => $"{attendee}: .NET Developer interested in AI";
var profileExecutor = getProfile.BindAsExecutor("GetAttendeeProfile");

Func<string, string> recommendSession =
 profile => profile.Contains("AI")
     ? "Building Agentic AI with Microsoft Foundry"
     : "What's New in .NET";
var recommendationExecutor = recommendSession.BindAsExecutor("RecommendSession");

Func<string, string> createMessage = session => $"Recommended session: {session}";
var messageExecutor = createMessage.BindAsExecutor("CreateRecommendation");

// 2. create MAF Workflow
var workflow = new WorkflowBuilder(profileExecutor)
  .AddEdge(profileExecutor, recommendationExecutor)
  .AddEdge(recommendationExecutor, messageExecutor)
  .WithOutputFrom(messageExecutor)
  .Build();

// 3. Run Workflow
await using Run run = await InProcessExecution.RunAsync(workflow, "Peter");
foreach (WorkflowEvent evt in run.NewEvents)
{
  if (evt is ExecutorCompletedEvent executorComplete)
  {
    Console.WriteLine($"{executorComplete.ExecutorId}: {executorComplete.Data}");
  }
}
```

# MICROSOFT AGENT FRAMEWORK

## AGENTS

- Runtime
- Agent Types
- Tools
- Middleware
- Conversations & Memory
- Safety

## WORKFLOWS

- Builder
- Executor
- Edges
- Events
- State Management
- Human-in-the-Loop

## HARNESS AGENT

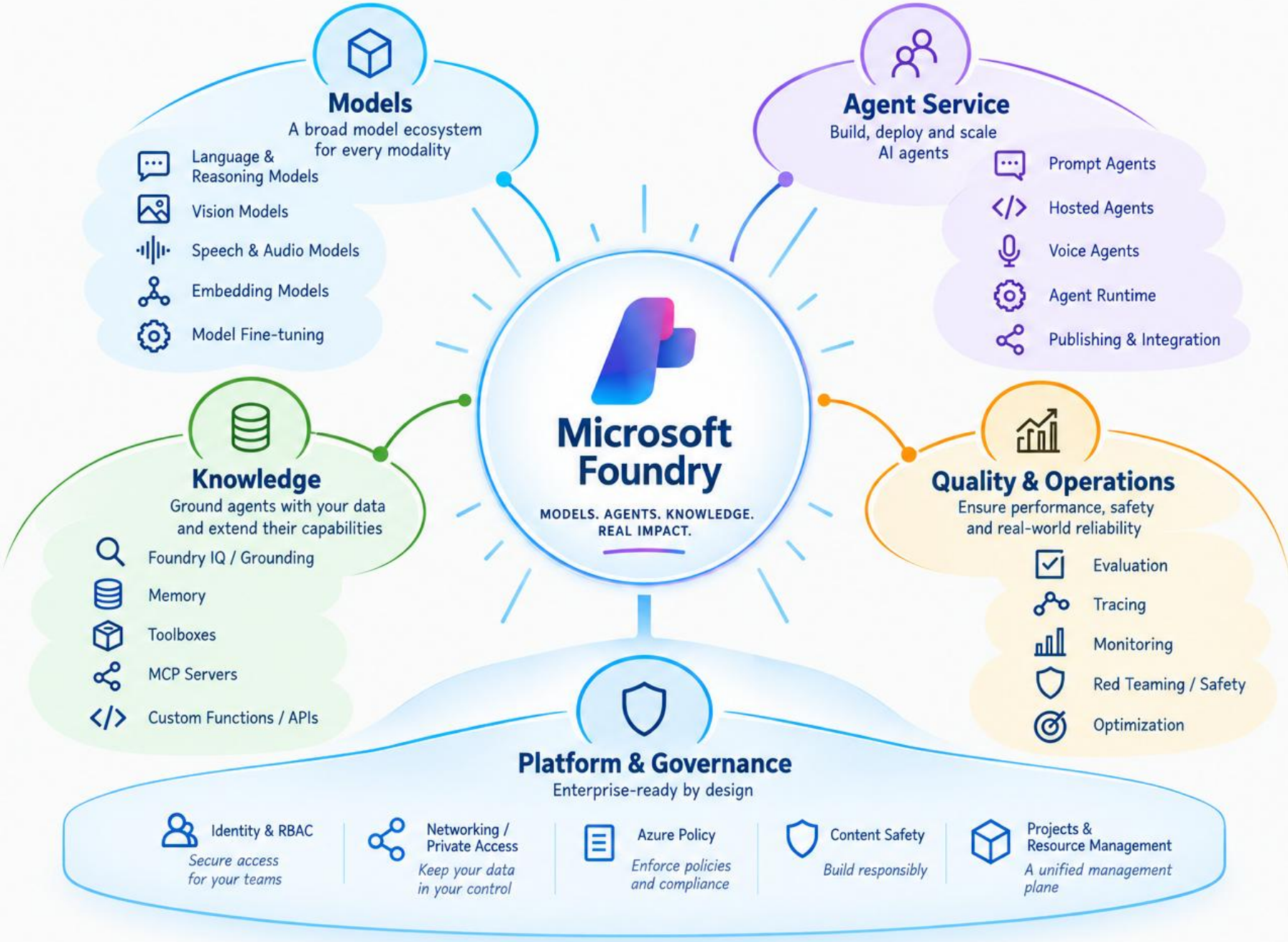
- Advanced Autonomy
- Agent Modes
- Compaction
- Todo Tracking
- Tools
- WebSearch
- Shell
- Looping
- Skills
- Background Agents

## INTEGRATIONS

- Microsoft Foundry
- Azure
- OpenAI
- Google
- Ollama
- Mistral
- TypeSafe AI
- Anthropic
- MCP
- A2A


C#

# MICROSOFT FOUNDRY


Models
A broad model ecosystem for every modality
Language & Reasoning Models
Vision Models
Speech & Audio Models
Embedding Models
Model Fine-tuning
Agent Service
Build, deploy and scale AI agents
Prompt Agents
Hosted Agents
Voice Agents
Agent Runtime
Publishing & Integration
Microsoft Foundry
MODELS. AGENTS. KNOWLEDGE. REAL IMPACT.
Knowledge
Ground agents with your data and extend their capabilities
Foundry IQ / Grounding
Memory
Toolboxes
MCP Servers
Custom Functions / APIs
Quality & Operations
Ensure performance, safety and real-world reliability
Evaluation
Tracing
Monitoring
Red Teaming / Safety
Optimization
Platform & Governance
Enterprise-ready by design
Identity & RBAC
Secure access for your teams
Networking / Private Access
Keep your data in your control
Azure Policy
Enforce policies and compliance
Content Safety
Build responsibly
Projects & Resource Management
A unified management plane


C#

# AGENTIC DESIGN PATTERNS

# AGENTIC DESIGN PATTERN FAMILIES

(APPLY ACROSS ALL LEVELS)

## FLOW

- Prompt Chaining
- Routing
- Parallelization
- Planning
- Prioritization
- Decision Models

ORCHESTRATE WORK EFFECTIVELY

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ACHIEVE MORE TOGETHER

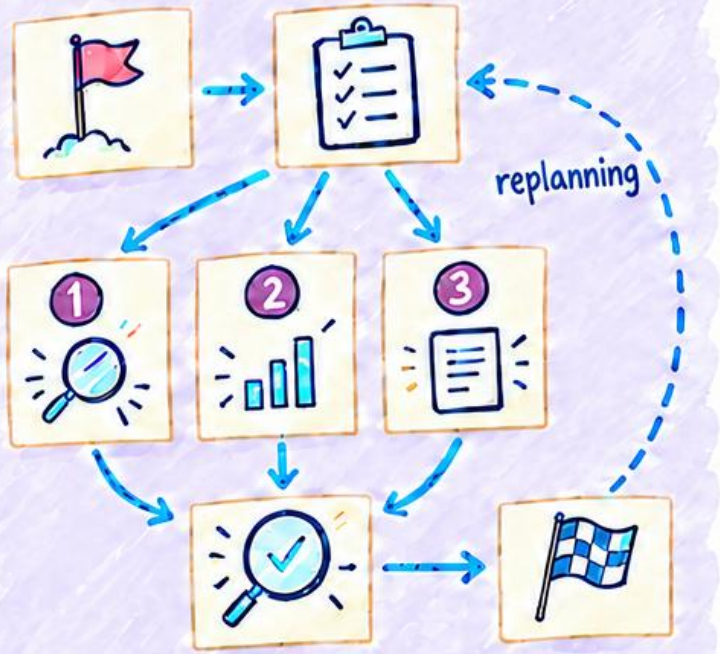
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- Exception Handling
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A SAFE, RELIABLE AND SUSTAINABLE FOUNDATION FOR AGENTS AT ANY SCALE

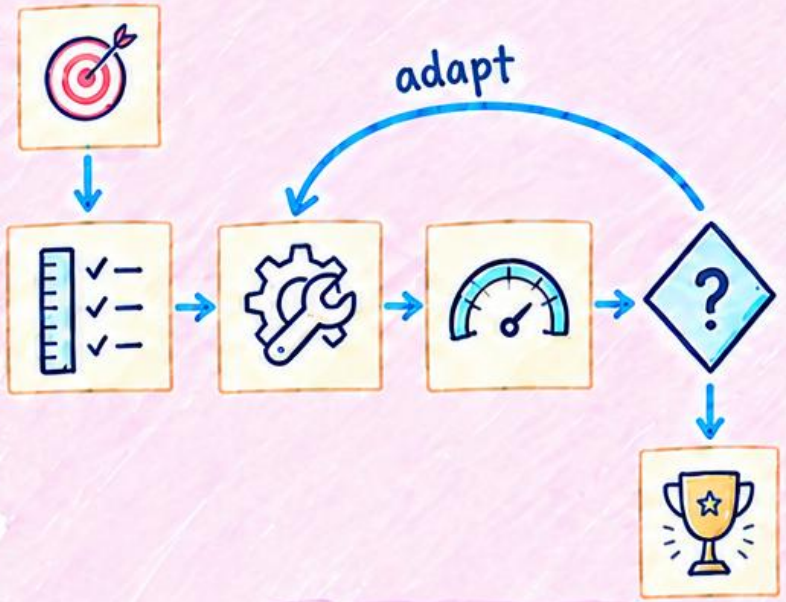
# PLANNING, GOAL MONITORING, REFLECTION

# PLANNING



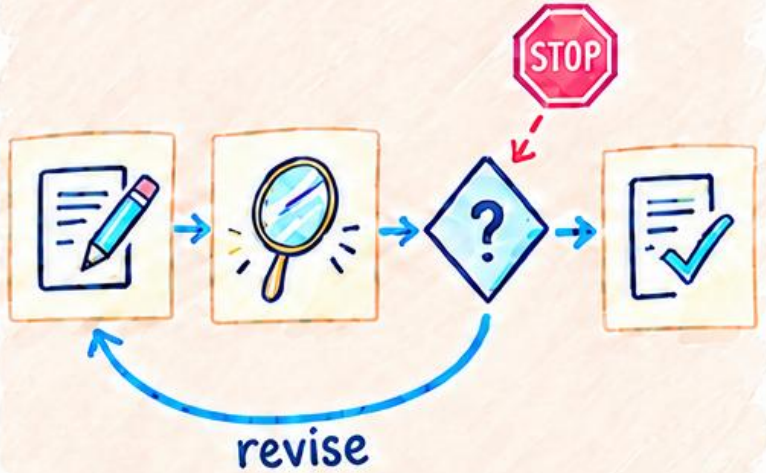

PLAN, ACT, ADAPT

# GOAL SETTING & MONITORING



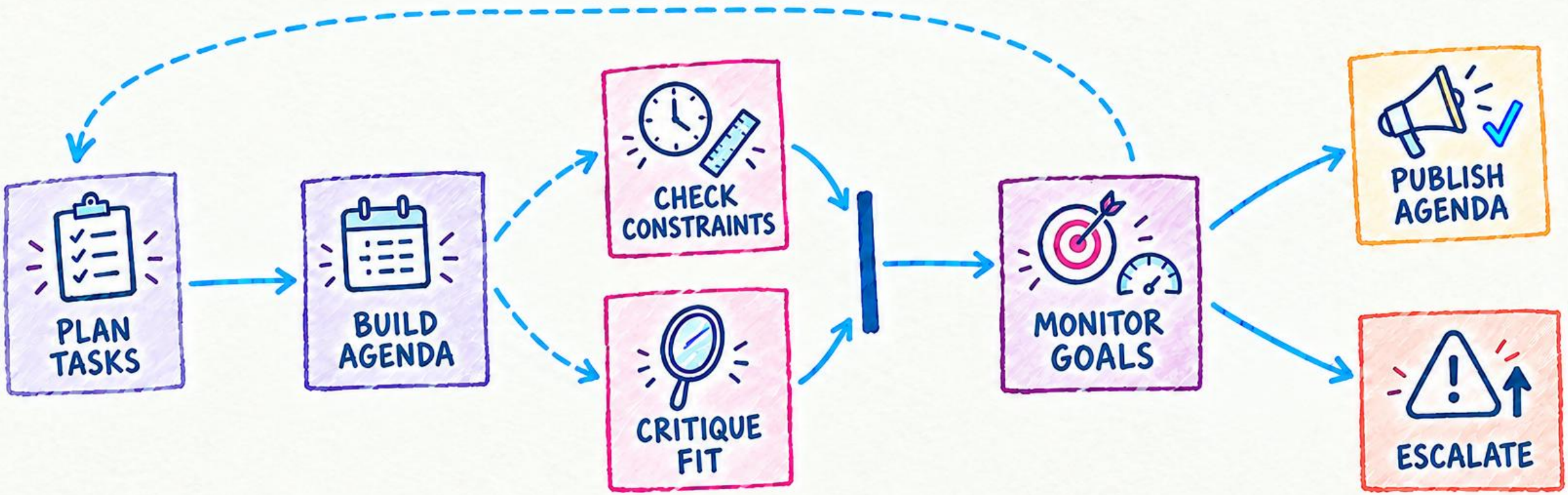

MEASURE PROGRESS, THEN ADAPT

# REFLECTION




CRITIQUE, REVISE – OR STOP

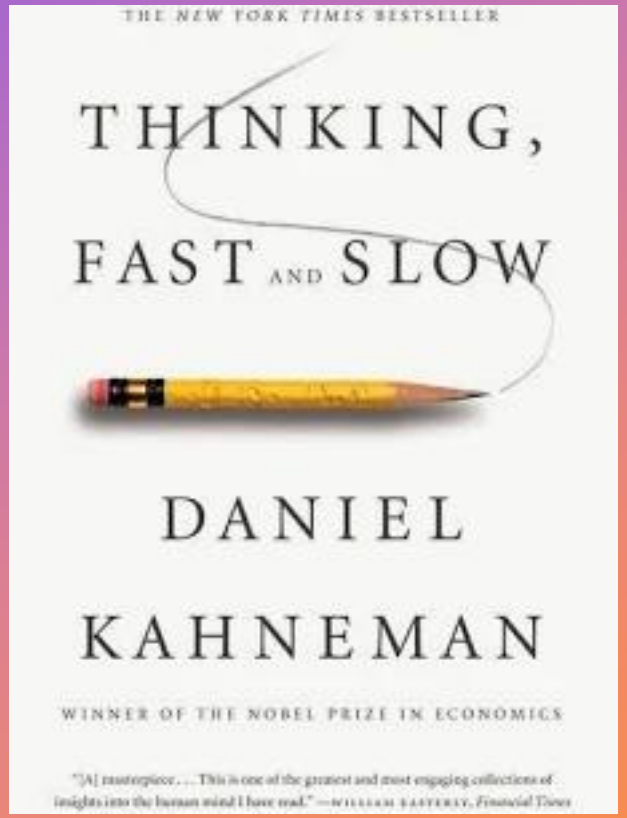
# UPDATE CONFERENCE PLANNER



```
// executors
var plan = new PlanTasks(client, verbose);
var build = new BuildAgenda(client, schedule);
var constraints = new CheckConstraintsExecutor(schedule);
var critic = new CritiqueFit(client, schedule);
var monitor = new MonitorGoals(verbose);
var publish = new PublishAgenda();
var escalate = new Escalate();

return new WorkflowBuilder(plan)
    .WithName("AgendaGoalMonitoring")
    .WithDescription("Task plan → build agenda → parallel review → goal-driven replanning.")
    .AddEdge(plan, build, "task strategy")
    .AddFanOutEdge(build, [constraints, critic], "draft agenda")
    .AddFanInBarrierEdge([constraints, critic], monitor, "goal checks")
    .AddEdge<PlanningRound>(monitor, plan, round ⇒ round is not null, "goals unmet: revise")
    .AddEdge<AgendaPlanResult>(monitor, publish, result ⇒ result is { Achieved: true }, "all goals met")
    .AddEdge<AgendaPlanResult>(monitor, escalate, result ⇒ result is { Achieved: false }, "budget exhausted")
    .WithOutputFrom(publish, escalate)
    .Build();
```

# DECISION MODELS



# LLM VS. DECISION MODEL

## SYSTEM 2 · LLM



THINK ABOUT IT.

## SYSTEM 1 · DECISION MODEL




MAKE THE CALL.

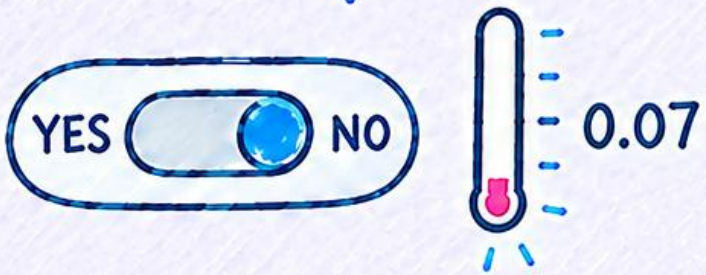
# DECISION MODEL

## NOUL (PREDICATE)

Binary Classification

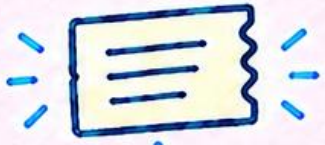


"Does this request concern a billing problem?"




## CHOICE

Multi-Class Classification



"Which team should handle this ticket?"

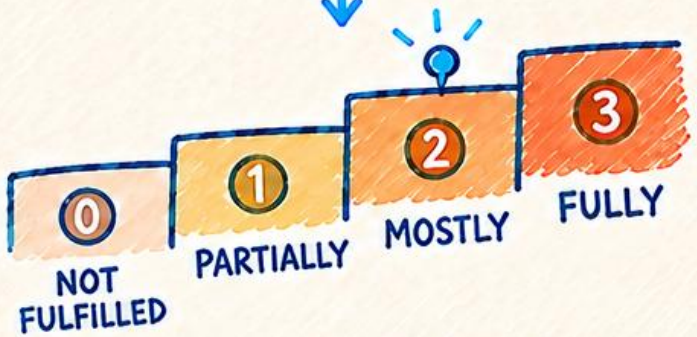



## SCORE

Ordinal Classification



"How completely does this answer satisfy the requirement?"



```
// 0. Decision API not yet part of MEAI
using ElBruno.AI.Decisions;

// 1. initialize client
var client =  new FoundryDecisionClient(new FoundryDecisionOptions
{
    Api = FoundryDecisionApi.SystemOne | FoundryDecisionApi.OpenAIDecisions,
    Endpoint = endpoint
    Model = "Decision-1" | "gpt-6-luna" ,
    ApiKey = apikey
});
```

```
// 2. Noul
var schedule = "" // the update conference schedule
var noulQuestion = "Is Cédric speaking at the event day 1?";
var noulResult = client.AssessAsync(schedule, noulQuestion);
// noulResult.Probability
```

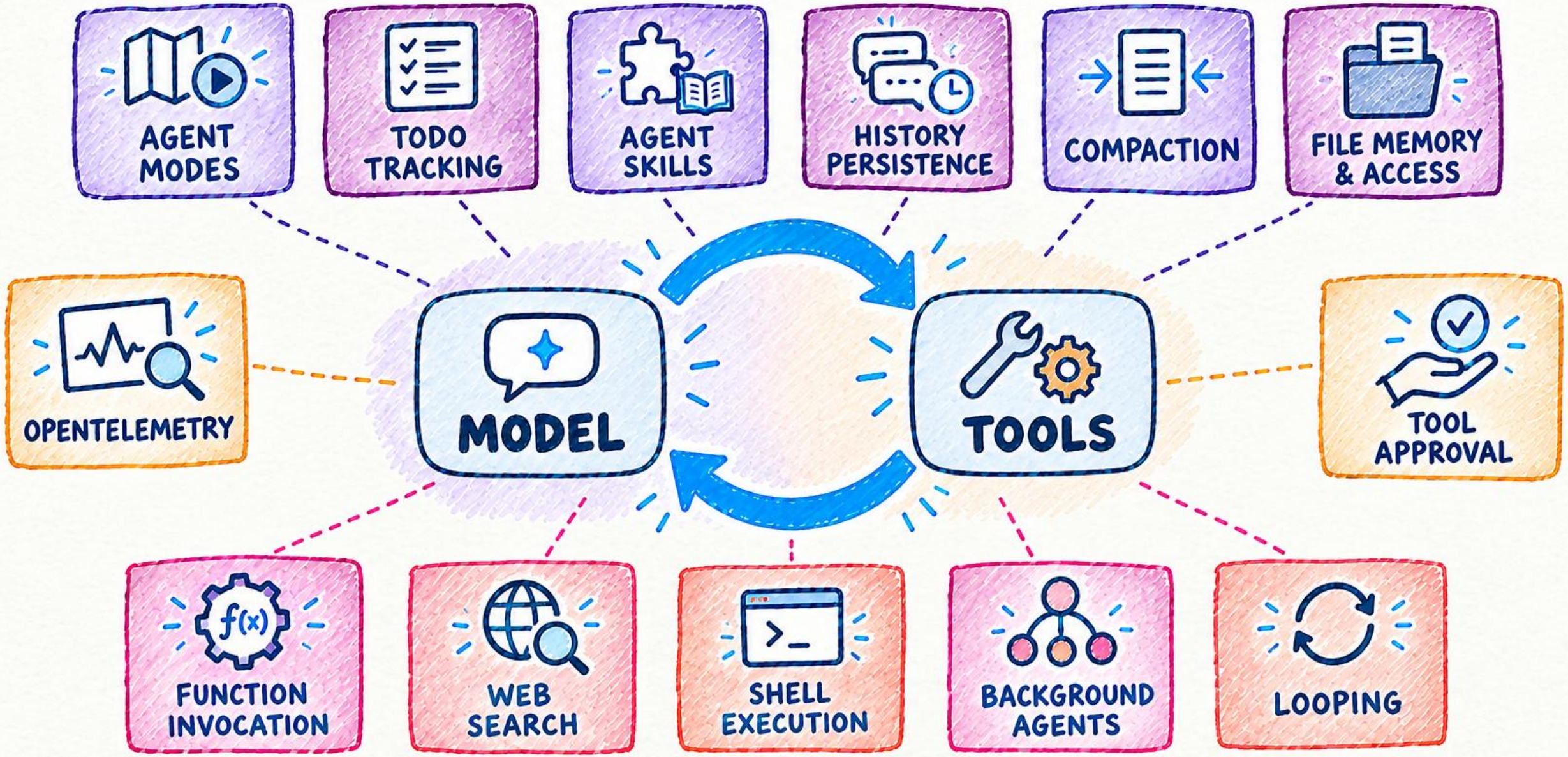
```
// 3. Choice
var ticketDescription = "";
var choiceQuestions = "Which team should handle this ticket?";
var teams = new ReadOnlyDictionary<string, string?>(new Dictionary<string, string?>
{
    ["billing"] = "Charges, invoices, payments, and refunds.",
    ["technical"] = "Software errors, bugs, failed integrations, and outages.",
    ["account"] = "Sign-in, password, and account-access problems.",
    ["unclear"] = "Insufficient evidence, multiple equally important problems, or outside these teams."
});
var choiceResult = await client.ChooseAsync(ticketDescription, choiceQuestions, teams);

// choiceResult.Choice, choiceResult.Confidence, choiceResult.Probabilities
```

```
// 4. score
var ticketDescription = "";
var scoreQuestion = "How severe is this incident?";
var levels = new[]
{
    "Minor: little user impact",
    "Moderate: some users affected; workaround available",
    "Major: many users affected or critical functionality unavailable",
    "Critical: business-critical service unavailable; no workaround"
};
var scoreResult = await client.ScoreAsync(ticketDescription, scoreQuestion, levels);
// scoreResult.Score, scoreResult.LevelProbabilities
```

# AGENT HARNESS

# AGENT HARNESS



```
public static HarnessAgent CreateAgent(IChatClient client, SpeakerProfile profile, string outputDirectory, bool webSearch) =>
client.AsHarnessAgent(new HarnessAgentOptions
{
    Name = "SpeakerPageBuilder",
    ChatOptions = new ChatOptions
    {
        Tools =
        [
            AIFunctionFactory.Create(ListSessions), AIFunctionFactory.Create(GetSession),
            AIFunctionFactory.Create((string username, CancellationToken ct) => DownloadGitHubAvatarAsync(username, outputDirectory, ct),
                "DownloadGitHubAvatar", "Download the public GitHub avatar of a GitHub username into the page folder"),
            AIFunctionFactory.Create(() => CheckPage(Path.Combine(outputDirectory, PageFile), profile, webSearch),
                "CheckPage", "Run the code checks on the saved page; fix every failed check and call again.")
        ],
        AllowMultipleToolCalls = false // one tool call per turn keeps the persisted tool history valid
    },
    AgentSkillsSource = new AgentFileSkillsSource(SkillsDirectory),  // how to build the page
    ToolApprovalAgentOptions = new ToolApprovalAgentOptions
    {
        AutoApprovalRules = [AgentSkillsProvider.ReadOnlyToolsAutoApprovalRule] // reading a skill is safe
    },
    FileAccessStore = new FileSystemAgentFileStore(outputDirectory), // the only folder it can write to
    FileAccessProviderOptions = new FileAccessProviderOptions
    {
        DisableReadOnlyToolApproval = true,
        DisableWriteToolApproval = true // the sandbox is the boundary
    },
    DisableWebSearch = !webSearch,  // hosted web_search tool, executed by Foundry
    DisableAgentModeProvider = true // just execute; no plan-then-approve stop
});
```

# CLOSING


C#

# AGENTIC DESIGN PATTERN FAMILIES

(APPLY ACROSS ALL LEVELS)

## FLOW

- Prompt Chaining
- Routing
- Parallelization
- Planning
- Prioritization
- Decision Models

ORCHESTRATE WORK EFFECTIVELY

## REASON & IMPROVE

- Reflection
- Learning & Adaptation
- Goal Setting & Monitoring
- Reasoning
- Exploration & Discovery

GET BETTER OVER TIME

## KNOWLEDGE & ACTION

- Tool Use
- Memory
- MCP
- RAG

BRIDGE IDEAS AND THE REAL WORLD

## COLLABORATION

- Multi-Agent Collaboration
- Human-in-the-Loop
- A2A




ACHIEVE MORE TOGETHER

## TRUST & OPERATIONS

- Exception Handling
- Resource Optimization
- Guardrails
- Evaluation & Monitoring
- Agent Harness

A SAFE, RELIABLE AND SUSTAINABLE FOUNDATION FOR AGENTS AT ANY SCALE

# START BUILDING AI APPLICATIONS IN C#

GITHUB – SLIDES AND DEMO

LINKEDIN

MEDIUM

#UCP26

Session: 13

# GIVE US FEEDBACK FOR THIS SESSION

Agentic AI on Microsoft Foundry: From Function Calling to Multi-Agent Systems

Update Conference
PRAGUE 2026

# Start building AI applications in C#



# Ressources

- GitHub Demo & Slides
- Agentic Design AI
- Azure Agent Landing Zone
- Microsoft Foundry – Docs
- Microsoft Agent Framework - Docs